LAB EXERCISE 1: File Handling (Student Marks)

Objective

To learn how to open, read, write, and close files in C++ while processing and analyzing student marks data.

Instructions

Based on the given Student.h and main.cpp, create Student.cpp so that the program performs the following tasks:

- Reads student data (name and marks) from a text file.
- Calculates the average mark.
- Displays the results in the console.
- Writes the results back into an output file.

Student.h

```
#ifndef STUDENT_H
#define STUDENT_H

#include <string>
#include <vector>

class Student {
public:
    Student(const std::string& name, double mark);
    const std::string& getName() const;
    double getMark() const;

private:
    std::string name;
    double mark;
};

class StudentManager {
public:
    bool loadFromFile(const std::string& filename);
    void displayAll() const;
    double calculateAverage() const;
    bool saveToFile(const std::string& filename) const;

private:
```

```
    std::vector<Student> students;
};

#endif
```

main.cpp

```
#include "Student.h"
#include <iostream>
using namespace std;

int main() {
    StudentManager manager;

    if (manager.loadFromFile("students.txt")) {
        cout << "File loaded successfully!\n";
    } else {
        cout << "Error: Unable to open input file.\n";
        return 1;
    }

    manager.displayAll();
    cout << "\nAverage mark: " <<
manager.calculateAverage() << endl;

    if (manager.saveToFile("results.txt")) {
        cout << "Results saved to file successfully!\n";
    } else {
        cout << "Error: Unable to save file.\n";
    }

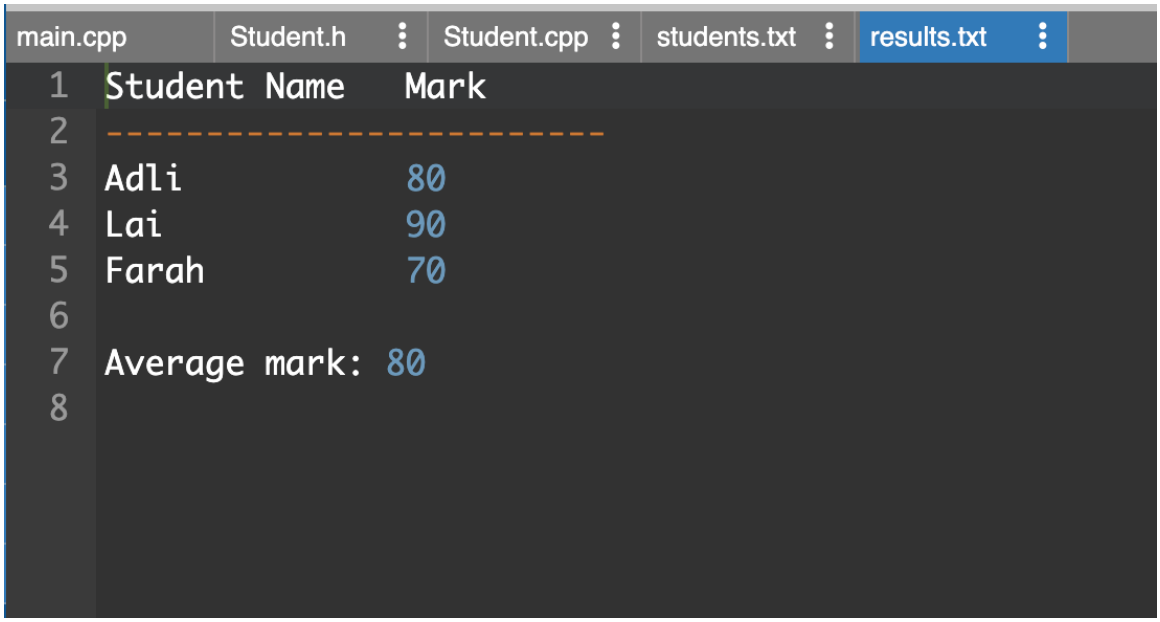
    return 0;
}
```

Expected Output

File loaded successfully!
Student Name Mark

Aiman 80
Lai Kin Yong 90
Farah 70

Average mark: 80
Results saved to file successfully!



Sample Input File (students.txt)

Adli 80
Lai 90
Farah 70

Your Task

Create Student.cpp so that:
• The program can open and read students.txt.
• It stores data into a vector<Student>.
• It displays all students with proper formatting.
• It calculates the average mark correctly.
• It writes the output (including average) to results.txt.